

Abstract

Modern enterprise information systems contain heterogeneous streams of information that differ fundamentally in representation, temporal structure, and semantics. Relational databases encode structured attributes and transactions; knowledge graphs encode entities and relations; textual systems encode sequential symbolic information; telemetry and event streams additionally encode temporal dynamics. Conventional memory systems typically transform these modalities into a common vector space and subsequently perform similarity search, graph traversal, or learned retrieval. Such approaches provide effective retrieval but do not naturally model one-shot associative learning, temporal binding, pattern completion, continual online learning, or biologically inspired memory consolidation.

This work proposes a **Polychronous Multimodal Episodic Memory (PMEM)** architecture based on a spiking neural substrate inspired by the entorhinal cortex (EC), dentate gyrus (DG), hippocampal CA3, and CA1. Heterogeneous modalities are independently transformed into temporally structured spike trains and converged into a common entorhinal representation. The DG performs sparse pattern separation, while recurrent CA3 dynamics form autoassociative attractor states capable of completing memories from partial or cross-modal cues. Spike-timing-dependent plasticity (STDP), neuromodulatory gating, synaptic delays, and polychronous convergence provide mechanisms for rapid associative binding. CA1 provides a learned readout from the latent hippocampal state back to modality-specific representations. Subsequent extensions introduce episodic temporal sequences, replay-based consolidation, continual learning, homeostatic plasticity, modular scaling, and cross-modal retrieval.

The resulting system is not intended to replace relational databases, graph databases, or language models. Instead, it defines a **neural associative memory substrate operating over these systems**, in which structured records, graph relations, and textual episodes become temporally encoded neural events. The central research hypothesis is that multimodal information can be bound through spike timing and recurrent dynamics without requiring a single shared semantic embedding or an explicit retrieval index. The current prototype demonstrates encoder reconstruction, CA3 pattern completion, multimodal binding, engram stability, novelty-sensitive plasticity, and continuous two-stage encoding; however, several components remain prototype-level, particularly pure neural retrieval without symbolic assistance, large-scale capacity, semantic text encoding, continual consolidation, and rigorous comparison against modern memory baselines.

spiking_multimodal_memorynew

1. Motivation

1.1 Problem Definition

Enterprise information is intrinsically heterogeneous.

A single real-world event may simultaneously appear as:

structured state + relational structure + language + temporal context

For example:

SQL:

```
employee_id = 18291
employee = Alice
department = AI
salary = 85000
company = Google
```

Graph:

```
Alice —WORKS_AT—> Google
Alice —REPORTS_TO—> Bob
```

Text:

```
"Alice joined Google's AI division in 2026
and now reports to Bob."
```

Time:

```
2026-08-11 14:32:08
```

These representations describe the **same underlying episode**, yet conventional systems normally process them independently.

A relational database provides exact lookup and aggregation. A graph database provides relational traversal. A vector database provides approximate semantic retrieval. A language model provides high-dimensional learned representations. None of these mechanisms, by themselves, directly implement a biologically inspired mechanism in which a partial cue dynamically reactivates a distributed multimodal episode.

The proposed system treats these modalities as analogous to sensory streams entering a common associative memory.

The conceptual transformation is:

heterogeneous information → spatiotemporal spike patterns → associative n



rather than:

heterogeneous information → single vector → k-NN retrieval

The original engineering specification defines the central claim as a spiking network using delay-heterogeneous synapses and STDP to bind heterogeneous modalities into unified episodic memories without requiring explicit synchronization or a shared embedding space.

SPIKING_MEMORY_SPEC

2. Research Motivation

The architecture is motivated by five unresolved computational problems.

2.1 Multimodal binding

How can independently encoded SQL, graph, and text information become one memory rather than three unrelated representations?

2.2 Pattern separation

How can highly similar episodes avoid being represented by nearly identical neural states?

For example:

```
Alice → Google
Alice → Microsoft
Bob   → Google
Bob   → Microsoft
```

must produce sufficiently distinguishable memory representations.

2.3 Pattern completion

Given:

```
Alice → Google
```

the system should reconstruct:

```

Alice
salary = 85000
department = AI
Google
WORKS_AT
textual description
temporal context

```

without explicitly storing a conventional lookup table for the answer.

2.4 Continual learning

The system should learn:

$$E_1, E_2, \dots, E_N$$

sequentially without requiring complete retraining.

2.5 Temporal association

Information arriving at different times should still be bound when its temporal relationship is informative.

The central hypothesis is therefore:

Spike timing, synaptic delays, sparse population coding, recurrent attractor dynamics, and local plasticity can form a computational substrate for heterogeneous episodic memory.

This research direction was previously identified as a core problem: designing a universal spiking computation capable of ingesting radically different modalities through local learning and dynamical attractor states.

Pasted markdown(5)

3. Terminology and Computational Interpretation

3.1 Spiking Neural Network

A spiking neural network (SNN) represents information using discrete events:

$$s_i(t) = \sum_k \delta(t - t_i^k)$$

where t_i^k denotes the time of the k -th spike from neuron i .

Information can therefore exist in:

- which neurons fire,
- how many times they fire,
- when they fire,
- relative timing between spikes,
- propagation delays,
- collective population activity.

3.2 Population Coding

A scalar value is represented by a group of neurons rather than one neuron.

For a continuous variable such as age:

$$x = 30$$

multiple neurons have preferred values:

$$c_1, c_2, \dots, c_n$$

and their firing rates may follow:

$$r_i(x) = r_{\max} \exp \left(- \frac{(x - c_i)^2}{2\sigma^2} \right)$$

Thus:

Age = 30

Neuron 1	low
Neuron 2	low
Neuron 3	medium
Neuron 4	HIGH
Neuron 5	HIGH
Neuron 6	medium
Neuron 7	low
...	

The population jointly represents the value.

The prototype uses 20 neurons per SQL field and reconstructs numeric values from the population response. SPIKING_MEMORY_SPEC

3.3 Modality Encoder

A modality encoder converts a non-neural representation into a spike representation.

The architecture contains:

$$E_{SQL}, E_{Graph}, E_{Text}, \dots$$

such that:

$$x_{SQL} = E_{SQL}(d_{SQL})$$

$$x_{Graph} = E_{Graph}(d_{Graph})$$

$$x_{Text} = E_{Text}(d_{Text})$$

The encoder does **not** need to make all modalities identical. It only needs to convert them into compatible temporal spike events.

3.4 Graph Population Coding

A graph node is represented by a sparse ensemble.

For example:

```
Alice
  ↓
20 neurons
  ↓
5 active neurons
```

The prototype specification uses a 20-neuron node ensemble with 5 active neurons.

SPIKING_MEMORY_SPEC

3.5 Synaptic Delay

A spike transmitted through a synapse does not necessarily arrive immediately.

If:

$$t_{pre} = 2ms$$

and:

$$d = 5ms$$

then:

$$t_{arrival} = 7ms$$

Synaptic delay therefore becomes an information-bearing variable.

3.6 Polychronization

Polychronization refers to the formation of temporally coordinated neuronal groups in which spikes emitted at different times arrive at downstream neurons within a narrow coincidence window because of heterogeneous synaptic delays.

For example:

SQL spike:

t = 0 ms

delay = 7 ms

arrival = 7 ms

Graph spike:

t = 3 ms

delay = 4 ms

arrival = 7 ms

Text spike:

t = 5 ms

delay = 2 ms

arrival = 7 ms

Although the modalities were generated at different times:

$$0, 3, 5ms$$

their downstream arrivals coincide:

$$7, 7, 7ms$$

This creates a potential **polychronous convergence event**.

The earlier prototype experiments explicitly demonstrated delay-induced simultaneous arrival and STDP-dependent strengthening.

Pasted markdown(20260807-144408)

3.7 Entorhinal Cortex — EC

EC is treated as the gateway between modality-specific representations and hippocampal memory.

Its computational roles are:

1. multimodal convergence,
2. novelty estimation,
3. contextual gating,
4. modulation of plasticity.

The prototype computes a novelty-related energy and uses it to determine whether stronger plasticity should occur.

spiking_multimodal_memorynew

3.8 Novelty

Novelty measures how different the current event is from the system's learned state.

Conceptually:

$$E = \|x - Wh\|^2$$

where:

- x = current multimodal input,
- W = learned EC representation,
- h = current latent representation.

A large prediction error indicates that the current event is poorly explained by existing representations.

3.9 Acetylcholine / Neuromodulatory Gate

Novel events receive stronger plasticity.

Conceptually:

$$M = \begin{cases} 2, & \text{novel} \\ 1, & \text{familiar} \end{cases}$$

The prototype uses $M = 2$ for novel episodes and $M = 1$ for familiar episodes.

spiking_multimodal_memorynew

The intended biological interpretation is an acetylcholine-like signal that temporarily increases the probability of memory formation.

3.10 Dentate Gyrus — DG

DG performs **pattern separation**.

Suppose:

Episode A:
Alice + Google

Episode B:
Alice + Microsoft

Their raw representations may overlap substantially.

DG transforms them into sparse patterns:

A:
001000000010000100...

B:
000010001000000001...

The goal is:

$$\textit{Similarity}(DG(A), DG(B)) < \textit{Similarity}(\textit{Input}(A), \textit{Input}(B))$$

The specification describes DG as a sparse separator and proposes learned sparse coding.

SPIKING_MEMORY_SPEC

3.11 Sparse Coding

A future research-grade DG can solve:

$$\min_h \|x - Wh\|_2^2 + \lambda \|h\|_1$$

The first term preserves information.

The second term forces sparsity.

An additional orthogonality term can encourage different memories to occupy distinct regions:

$$\mathcal{L} = \|x - Wh\|_2^2 + \lambda \|h\|_1 + \mu \sum_{i < j} (h_i^T h_j)^2$$

This has been proposed as a route toward adaptive pattern separation.

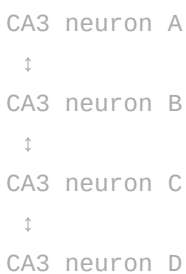
Pasted markdown(20260807-144408)

3.12 CA3

CA3 is the central associative memory.

Unlike DG, whose primary computational purpose is separation, CA3 uses recurrent connections to create **autoassociation**.

Conceptually:



The connections among these neurons become strengthened when they repeatedly participate in the same episode.

3.13 Attractor Network

An attractor network contains stable states toward which network activity naturally evolves.

Suppose an episode produces:

A B C D E

The corresponding CA3 state becomes an attractor:

$$\mathbf{x}_{ABC...E}$$

A partial cue:

A C E

may cause:

A C E

↓

recurrent excitation

↓

B

↓

D

↓

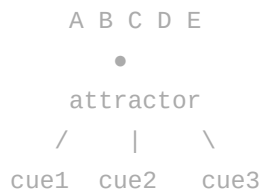
full assembly

The network has therefore completed the memory through its own dynamics.

3.14 Attractor Basin

The **attractor basin** is the set of initial states that converge toward the same attractor.

Conceptually:



If:

partial cue $\in$ basin

then:

$$x(t) \rightarrow x^*$$

where x^* is the stored memory state.

A good memory system should maximize the basin size without causing neighboring memories to collapse into the same basin.

3.15 Engram

An engram is the distributed neural substrate associated with a memory.

In this architecture:

$$Engram_i = \{n_1, n_2, \dots, n_k\}$$

where each n_i is a CA3 neuron participating strongly in episode i .

The engram is therefore not a single neuron or a stored object.

It is:

distributed neurons + strengthened synapses + temporal structure

The hippocampal engram is commonly conceptualized as an index for distributed memory components.

Pasted markdown(6)

3.16 STDP

Spike-timing-dependent plasticity changes a synaptic weight according to the temporal relationship between pre- and postsynaptic spikes.

For a pre-before-post event:

$$\Delta t = t_{post} - t_{pre} > 0$$

a simplified potentiation rule is:

$$\Delta w = A_+ e^{-\Delta t / \tau_+}$$

For post-before-pre:

$$\Delta t < 0$$

a simplified depression rule is:

$$\Delta w = -A_- e^{\Delta t / \tau_-}$$

The specification derives repeated potentiation as:

$$w_{n+1} = w_n + A_+ e^{-\Delta t / \tau_+}$$

and notes that saturation occurs as the weight approaches w_{max} .

SPIKING_MEMORY_SPEC

3.17 Plasticity

Plasticity is the ability of the network to change its internal connectivity as a consequence of experience.

The proposed system contains several levels:

Fast:

STDP

Medium:

neuromodulated plasticity

Slow:

consolidation

Homeostatic:

activity regulation

Plain pairwise STDP is not expected to be sufficient for the complete research system; modulated STDP and potentially delay learning are identified as future extensions.

Pasted markdown(6)

3.18 CA1

CA1 acts as a readout layer.

The computational transformation is:

$$CA3 \rightarrow CA1 \rightarrow \{SQL, Graph, Text\}$$

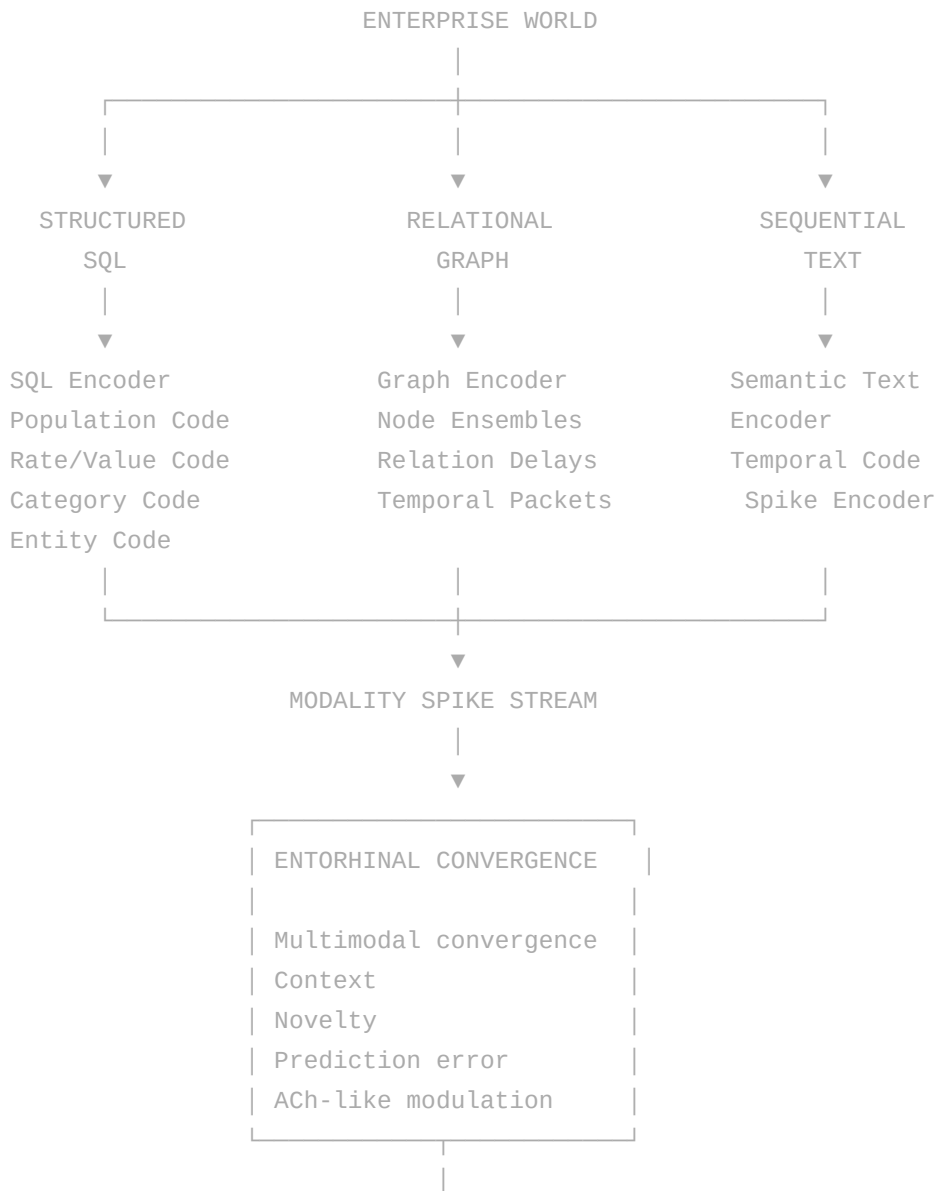
The current prototype already trains CA1 against SQL and graph support patterns.

spiking_multimodal_memorynew

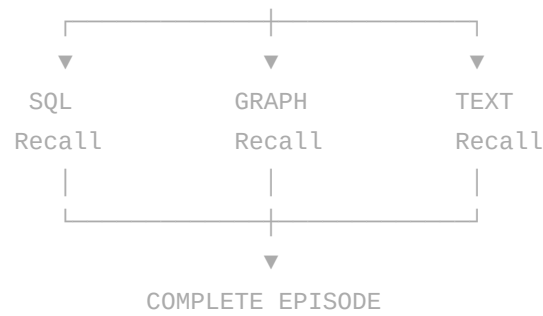
The research-grade version extends this to actual semantic reconstruction.

4. Complete Architecture

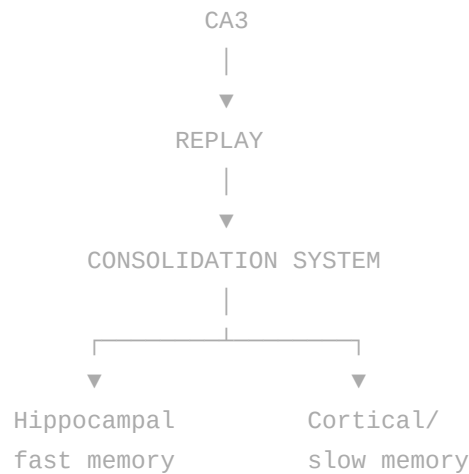
The final v0–v6 architecture is:







Around this core:



This dual-memory direction is proposed specifically to address catastrophic forgetting.

Pasted markdown(20260807-144408)

5. End-to-End Information Flow

5.1 Input Episode

Consider:

SQL:
Alice
age = 30
salary = 85000
department = AI

Graph:
Alice —WORKS_AT—> Google

Text:
"Alice works in Google's AI division."

The system treats these as one episode:

$$E = (E_{SQL}, E_{Graph}, E_{Text}, C, T)$$

where:

- C = context,
- T = temporal information.

6. SQL Encoding

6.1 Continuous fields

For:

$$age = 30$$

the population encoder computes firing rates around preferred age values.

For:

$$salary = 85000$$

the salary range is normalized and represented by another population.

The prototype specifies 20 neurons per field and reconstructs numerical values through tuning-curve decoding. SPIKING_MEMORY_SPEC

6.2 Categorical fields

For:

department = AI

a categorical population code is used.

The final system should avoid treating arbitrary categorical IDs as continuous numbers.

Thus:

Engineering
AI

Finance

HR

must not become:

1

2

3

4

with numerical distance interpreted semantically.

Instead:

$$\textit{category} \rightarrow \textit{sparselearnedcode}$$

6.3 Entity IDs

Entity identifiers represent another edge case.

For:

```
customer_id = 839291
```

population coding based directly on numeric magnitude is inappropriate.

The proposed final encoder therefore uses an entity code:

$$ID \rightarrow \textit{sparsehash/learnedentityrepresentation}$$

rather than:

$$839291 \rightarrow \text{Gaussian tuning curve}$$

7. SQL Edge Cases

Missing value

```
salary = NULL
```

must not become:

```
salary = 0
```

because zero and missing have different semantics.

Use:

```
value population
+
missingness population
```

Out-of-range value

If:

```
salary = 10,000,000
```

but the encoder expects:

$$0 \leq salary \leq 100,000$$

the system should not silently clip without recording the event.

Use:

```
overflow code
+
clipped population
```

New category

If:

```
department = Quantum Research
```

has never appeared:

```
unknown category
↓
novel entity/category allocation
```

↓
new sparse code

This should contribute to EC novelty.

Duplicate record

If the same event appears repeatedly:

Event A
Event A
Event A

the system should not allocate three independent memories.

Instead:

familiarity ↑

and plasticity should transition from rapid encoding toward reinforcement/consolidation.

8. Graph Encoding

Consider:

Alice —WORKS_AT—> Google

The source node has a sparse ensemble:

Alice:
N2 N7 N11 N15 N19

The target:

Google:
N1 N4 N8 N13 N17

The relation is encoded temporally.

For example:

$$d_{WORKS_AT} = 3ms$$

Thus:

$$t_{Google} = t_{Alice} + 3ms$$

The specification uses relation-specific delays such as $WORKS_AT = 3\text{ ms}$, $FRIENDS_WITH = 7\text{ ms}$, $MANAGES = 5\text{ ms}$, and $REPORTS_TO = 2\text{ ms}$. SPIKING_MEMORY_SPEC

9. Graph Edge Cases

Unknown node

Alice $\rightarrow$ WORKS_AT $\rightarrow$ UnknownCompany

A new entity code must be allocated rather than mapping the company to an existing node.

Same nodes, different relation

Alice $\xrightarrow{\text{WORKS_AT}}$ Google

Alice $\xrightarrow{\text{REPORTS_TO}}$ Google

must remain distinguishable.

Therefore relation information cannot depend solely on endpoints.

Same relation, different endpoints

Alice $\rightarrow$ WORKS_AT $\rightarrow$ Google

Bob $\rightarrow$ WORKS_AT $\rightarrow$ Microsoft

must also remain distinguishable.

Thus:

$$\text{memory} = \text{source} + \text{relation} + \text{target} + \text{timing}$$

Multiple edges

For:

Alice → WORKS_AT → Google

Alice → OWNS → Google

Alice → MANAGES → Google

the encoder must support multiple temporal packets rather than collapsing them.

10. Text Encoding

The text path is the least mature component of the current prototype and therefore receives a more substantial redesign in the final architecture.

The specification proposes a spiking tokenizer in which words are represented by sparse populations and temporal position is represented through phase-of-firing coding.

SPIKING_MEMORY_SPEC

The final architecture should use:

Text

↓

tokenization

↓

semantic representation

↓

temporal/event representation

↓

population code

↓

spike train

rather than sending raw words directly into CA3.

11. Text Encoding Example

Consider:

```
Alice joined Google's AI division.
```

The semantic encoder produces representations for:

```
Alice  
join  
Google  
AI division
```

Temporal coding preserves:

```
Alice  
↓  
joined  
↓  
Google  
↓  
AI division
```

The resulting spike stream might conceptually be:

```
t=0 ms    Alice  
t=4 ms    joined  
t=8 ms    Google  
t=12 ms   AI  
t=16 ms   division
```

The precise spike times are not merely timestamps; they become part of the neural representation.

12. Text Edge Cases

Synonyms

```
Alice works at Google.
```

```
Alice is employed by Google.
```

The semantic encoder should map these to partially overlapping representations.

Word order

Alice manages Bob.

Bob manages Alice.

must produce different temporal patterns.

Negation

Alice works at Google.

Alice does not work at Google.

must not produce nearly identical memories.

A dedicated compositional/negation code is therefore required.

Long documents

A 10,000-token document should not create 10,000 independent hippocampal events.

Instead:

```
document
  ↓
sections
  ↓
events / semantic units
  ↓
episodic packets
  ↓
hippocampal encoding
```

This prevents the CA3 network from becoming a token-level language model.

Repeated information

If a document repeatedly states:

Alice works at Google.

the system should consolidate the fact rather than create thousands of identical attractors.

Unknown vocabulary

Previously unseen words should not terminate processing.

They should receive:

subword/character representation
+
novelty signal

and subsequently acquire a learned representation.

13. Entorhinal Convergence

The modality outputs are concatenated or otherwise converged:

$$x = [x_{SQL}; x_{Graph}; x_{Text}]$$

The current implementation performs this explicitly before DG processing.

spiking_multimodal_memorynew

The final system extends this to:

$$x = [x_{SQL}; x_{Graph}; x_{Text}; x_{Time}; x_{Context}]$$

The purpose is not merely concatenation.

EC should estimate:

$$\delta_t = \|x_t - \hat{x}_t\|$$

and determine whether:

known event

or:

novel event

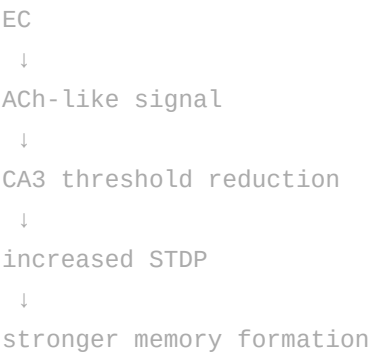
is being encountered.

14. Novelty → Neuromodulation

If:

$$\delta_t > \theta_N$$

then:



If:

$$\delta_t \leq \theta_N$$

then:

normal plasticity

The proposed implementation specifically suggests increasing STDP gain and lowering CA3 thresholds during novel events.

Pasted markdown(20260807-144408)

15. DG Processing

The converged representation enters DG:

$$x \in \mathbb{R}^D$$

and is transformed:

$$h = Wx$$

followed by sparse selection.

For example:

900 DG neurons

scores:

N1 = 0.01

N2 = 0.83

N3 = 0.04

...

N52 = 0.92

...

If:

$$k = 27$$

only the 27 highest-scoring neurons fire.

Thus:

900 candidates

↓

27 active

This converts a dense multimodal representation into a sparse neural code.

The current implementation uses a 900-neuron DG and top-k sparsification.

spiking_multimodal_memorynew

16. DG Edge Cases

Two nearly identical episodes

Desired:

$$J(DG(A), DG(B)) \ll J(Input(A), Input(B))$$

where J denotes Jaccard similarity.

Completely unrelated episodes

Desired:

$$DG(A) \cap DG(B) \approx \emptyset$$

Excessive similarity

If too many memories collapse into similar DG codes:

- increase dictionary capacity
- increase sparsity
- apply orthogonality regularization
- increase competitive inhibition

DG saturation

If too many neurons are active:

- homeostatic inhibition
- top-k enforcement
- adaptive sparsity

should restore sparse coding.

17. CA3 Memory Formation

The sparse DG pattern is projected into CA3.

The current implementation explicitly separates:

- SQL: 0–99
- Graph: 100–179
- Text: 180–279
- DG: 300–1199

before connecting these pathways into CA3.

This avoids the identifier collisions present in the earlier prototype.

18. Two-Stage Encoding

The current system uses:

Stage 1 — pre-activation

$$T_1 = 10ms$$

with:

$$M = 0$$

No major plasticity is intended.

The network determines:

Does this resemble something already stored?

Stage 2 — plasticity

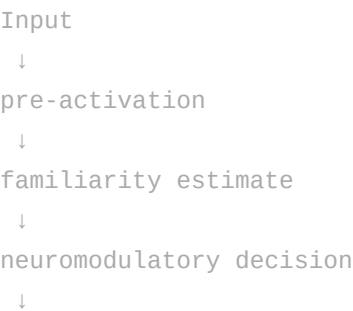
$$T_2 = 50ms$$

with:

$$M = \begin{cases} 2, & \text{novel} \\ 1, & \text{familiar} \end{cases}$$

The state is **not reset** between the two stages.

This creates:



continued dynamics

↓

STDP

↓

engram

19. CA3 Attractor Formation

Suppose the network selects:

CA3:

12

37

49

81

102

...

for an episode.

During learning, STDP strengthens connections among neurons participating in the same temporal pattern:

$$w_{ij} \rightarrow w_{ij} + \Delta w_{ij}$$

The resulting recurrent subnetwork becomes an attractor.

20. Polychronous Memory Formation

Assume:

SQL spike:

t=0

delay=7

Graph spike:

t=3

delay=4

Text spike:

$t=5$

$\text{delay}=2$

Then:

$$0 + 7 = 3 + 4 = 5 + 2 = 7ms$$

The downstream neuron receives three temporally aligned events.

If its membrane potential crosses threshold:

$$V(t) \geq V_{th}$$

it fires.

Then STDP strengthens the causal timing relationships.

This creates a temporally structured memory rather than a static binary pattern.

The original specification identifies heterogeneous delays as a mechanism for polychronous group formation.

SPIKING_MEMORY_SPEC

21. Pure Neural Retrieval

The final architecture must eliminate:

```

query
  ↓
signature lookup
  ↓
engram ID
  ↓
explicit engram injection

```

as the primary retrieval mechanism.

The target mechanism is:

```

partial cue
  ↓
encoder
  ↓
EC
  ↓

```

DG
↓
CA3 partial activation
↓
recurrent excitation
↓
competition
↓
attractor convergence
↓
complete CA3 engram
↓
CA1
↓
multimodal reconstruction

The distinction is fundamental.

A database lookup answers:

Which memory index corresponds to this query?

A CA3 attractor answers:

Which stable network state does this partial activity naturally converge toward?

The earlier simulation work explicitly identified recurrent amplification as necessary for genuine pattern completion.

Pasted markdown(20260807-144408)

22. Attractor Retrieval Example

Stored:

Memory M1:

Alice
Google
WORKS_AT
AI
salary 85k

CA3 engram:

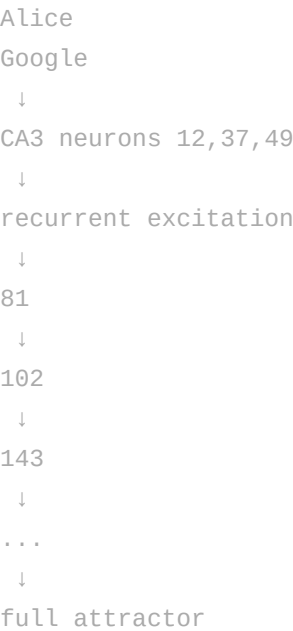
$\{12, 37, 49, 81, 102, 143, 177, \dots\}$

Query:

Alice + Google

Only part of the engram is activated.

Then:



The network converges to:

$$M_1$$

23. CA1 Reconstruction

Once CA3 settles:

$$r_{CA3}$$

is passed through CA1:

$$r_{CA1} = W_{CA1}r_{CA3} + b$$

The current prototype uses a trainable CA1 mapping and trains it against SQL and graph support patterns.

spiking_multimodal_memorynew

The final system extends this to:

$$CA3 \rightarrow \left\{ \begin{array}{l} SQL \\ Graph \\ Text \\ Temporalcontext \end{array} \right.$$

24. Cross-Modal Retrieval

This is one of the central final capabilities.

SQL → Graph

SQL:
Alice
department=AI



CA3 attractor



Graph:
Alice → WORKS_AT → Google

Graph → SQL

Alice → Google
↓
CA3
↓
salary=85k
department=AI

Text → Graph

"Alice joined Google's AI division."

↓

CA3

↓

Alice → WORKS_AT → Google

Text → SQL

"Alice works in Google's AI division."

↓

CA3

↓

department = AI
company = Google

Partial multimodal query

Alice
+
"works at Google"

should retrieve:

SQL
+
Graph
+
Text
+
temporal context

This is the target definition of **multimodal episodic binding**.

25. Episodic Extension

The final system must move beyond isolated facts.

Consider:

t1:

Alice joins Google.

t2:

Alice becomes manager.

t3:

Alice moves to Microsoft.

t4:

Alice becomes CTO.

The memory is:

$$M = (E_1, E_2, E_3, E_4)$$

rather than:

$$M = \{E_1, E_2, E_3, E_4\}$$

because ordering is part of the memory.

The temporal system therefore represents:

$$E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4$$

through sequential spike packets and recurrent temporal dynamics.

26. Temporal Retrieval

Query:

What happened before Alice became CTO?

The desired retrieval is:

$$E_3$$

rather than merely retrieving all facts involving Alice.

This requires the network to preserve:

$$t_i < t_j$$

relationships.

Temporal order is therefore encoded as a first-class memory variable.

27. Continual Learning

Enterprise information arrives continuously:

$$E_1, E_2, \dots, E_N$$

The system must learn online:

$$W_{t+1} = W_t + \Delta W_t$$

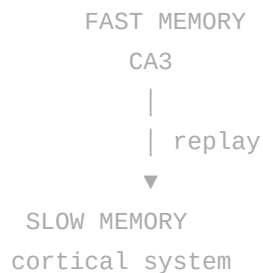
without restarting training.

This introduces three major problems:

1. catastrophic forgetting,
2. synaptic saturation,
3. memory interference.

28. Memory Consolidation

The proposed solution is a two-timescale memory system:



During low-input periods:

sharp-wave/ripple-like event

↓

CA3 reactivation

↓

old + recent memory replay

↓

slow cortical learning

↓

long-term consolidation

Replay-based consolidation has explicitly been proposed as a mechanism for preventing catastrophic forgetting. Pasted markdown(5)

29. Homeostatic Plasticity

STDP alone can cause unstable behavior.

If neuron i fires too frequently:

$$\theta_i \uparrow$$

If it rarely fires:

$$\theta_i \downarrow$$

A simple rule is:

$$\theta_i(t+1) = \theta_i(t) + \eta(\bar{r}_i - r_{target})$$

This creates more uniform allocation of memory participation.

Pasted markdown(20260807-144408)

30. Memory Interference

Suppose:

M1:

Alice → Google

M2:

Alice → Microsoft

M3:

Alice → Amazon

All three share:

Alice

A successful DG should separate them.

CA3 should store them as distinct attractors.

The experiment should measure:

$$J(M_i, M_j)$$

and retrieval accuracy as the number of memories increases.

31. Memory Capacity

The research specification proposes a polychronous capacity model of the form:

$$\mathcal{C} \sim N \binom{N}{k} \left(\frac{D_{max}}{\delta t} \right)^{k-1}$$

where:

- N = number of neurons,
- k = participating neurons,
- D_{max} = maximum delay,
- δt = temporal precision. SPIKING_MEMORY_SPEC

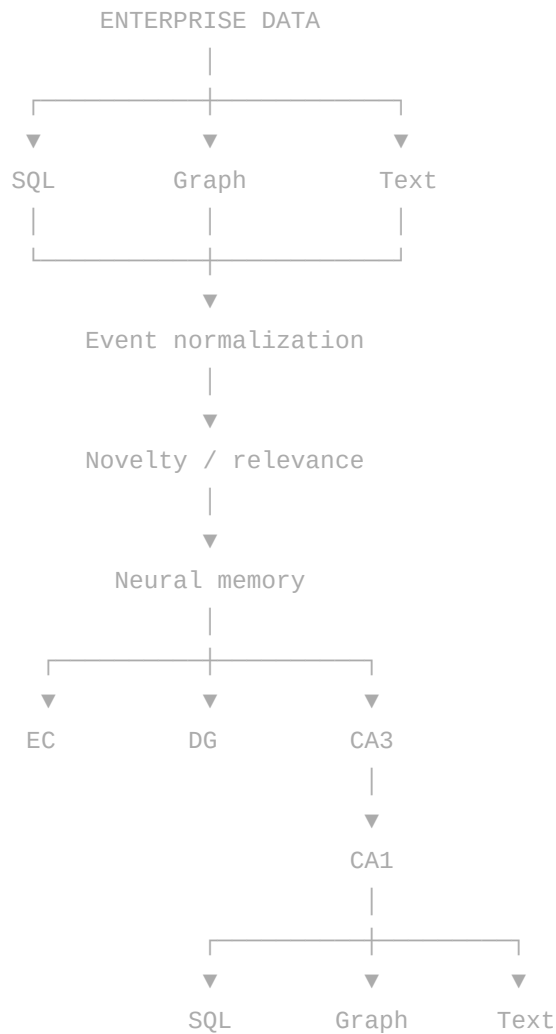
This should be treated as a **research hypothesis/model**, not as an established enterprise capacity guarantee.

Empirical capacity must ultimately be measured.

32. Enterprise-Scale Architecture

The final system should not attempt to replace enterprise databases.

Instead:



Conventional infrastructure continues to provide:

- exact transactions,
- durable storage,
- indexing,
- consistency,
- auditability,
- deterministic queries.

The spiking memory provides:

- associative recall,
- temporal binding,
- one-shot learning,
- pattern completion,
- multimodal association,
- continual adaptation.

33. V0–V6 Evolution

V0 — Neural Primitive

LIF neurons
Delayed synapses
STDP
Spike timing

Objective:

Establish that spike timing can modify synapses and produce temporal coincidence.

V1 — Working Hippocampal Prototype

SQL encoder
Graph encoder
EC
DG
CA3
CA1

Current prototype results include:

- encoder reconstruction: PASS,
- graph delay reconstruction: PASS,
- CA3 pattern completion: 10/10,
- engram stability: 0.91 mean Jaccard,
- novelty detection: PASS,
- continuous two-stage state: PASS.

spiking_multimodal_memorynew

34. V2 — Pure Neural Memory

Remove:

signature lookup
explicit engram injection
symbolic retrieval

Add:

learned DG→CA3
competitive engram formation
strong recurrent amplification
homeostatic allocation

Success criterion:

retrieval = CA3 dynamics only

This is the most important scientific transition.

35. V3 — Genuine Multimodal Memory

Replace simplistic text coding with semantic spiking encoding.

Add:

SQL ↔ Graph
SQL ↔ Text
Graph ↔ Text
SQL ↔ Graph ↔ Text

retrieval experiments.

The central question becomes:

Can one modality retrieve another without a modality-specific lookup table?



36. V4 — Episodic Memory

Add:

event sequences
temporal context
order
duration
contextual state

The memory becomes:

$$M = (E_1, E_2, \dots, E_T)$$

rather than an unordered set.

37. V5 — Continual Memory

Add:

100K-millions of episodes
interference
forgetting
replay
consolidation
homeostasis

The primary experiment becomes:

$$Accuracy(N)$$

as:

$$N = 10, 100, 1000, 10^4, 10^5, \dots$$

38. V6 — Research System

V6 adds the experimental infrastructure required for scientific validation:

biological constraints
+
ablation studies
+
baselines
+
capacity analysis
+
scaling
+
statistical testing
+
neuromorphic implementation

The target is no longer simply:

“Does the network work?”

but:

“Which mechanism produces which computational advantage, under what conditions, and at what cost?”

39. Major Novelty Claims

The following constitute the strongest potential research contributions.

Novelty 1 — Modality-independent spiking substrate

SQL, graph, text, and potentially future modalities use independent encoders but converge into a common spiking memory substrate.

The architecture therefore avoids forcing all modalities into one conventional embedding space.

Novelty 2 — Polychronous multimodal binding

The central hypothesis is that asynchronous modalities can become associated through:

relative spike timing + heterogeneous synaptic delays

rather than explicit synchronization.

The research specification explicitly identifies delay-heterogeneous convergence as the binding mechanism. SPIKING_MEMORY_SPEC

40. Novelty 3 — Dynamical rather than indexed retrieval

The target retrieval mechanism is:

query → *networkstate* → *attractor*

rather than:

query → *index* → *memory*

This distinction is essential to the core-AI interpretation of the project.

Pasted markdown(5)

41. Novelty 4 — Multimodal episodic engrams

One CA3 attractor should simultaneously bind:

```
structured attributes
+
graph relations
+
language
+
temporal context
```

The memory is therefore an **episodic multimodal engram**, rather than a vector representing only one modality.

42. Novelty 5 — Local online learning

The intended system learns using local mechanisms:

$$\Delta w = f(t_{pre}, t_{post}, M)$$

rather than requiring global backpropagation through the entire memory.

The research objective is one-shot or few-shot memory formation.

43. Novelty 6 — Novelty-driven memory allocation

EC determines whether an event should cause stronger plasticity.

Thus:

```
familiar:
weak update
```

novel:
strong update

This creates an adaptive memory allocation mechanism.

44. Novelty 7 — Temporal information as computation

In conventional neural networks, temporal information is often converted into vectors.

Here:

time itself becomes part of the representation

through:

- spike timing,
- synaptic delays,
- phase coding,
- temporal sequences,
- polychronous convergence.

45. Novelty 8 — Continual memory through replay

The system is designed to evolve from:

fast hippocampal memory

toward:

fast hippocampal memory
+
slow cortical memory
+
replay

This provides a route toward long-lived continual learning rather than indefinite STDP updates.

46. Major Problems and Proposed Solutions

Problem 1 — Encoding heterogeneous modalities

Problem

SQL, graphs, and text do not naturally share a representation.

Proposed solution

Independent modality encoders:

$$E_m : D_m \rightarrow S_m$$

followed by spike-based convergence.

Remaining research question

Does the encoding preserve semantic similarity and compositional structure?

This is explicitly identified as a research problem in the earlier design.

Pasted markdown(5)

47. Problem 2 — High-cardinality enterprise data

Problem

Enterprise databases contain:

```
billions of IDs
millions of categories
NULLs
JSON
timestamps
schema changes
```

A fixed 20-neuron population code is insufficient.

Proposed solution

Use heterogeneous encoders:

```
continuous:
population code
```

categorical:
sparse learned code

entity ID:
hash/learned code

timestamp:
temporal code

missing:
explicit missingness code

JSON:
hierarchical encoder

48. Problem 3 — Text scaling

Problem

Token-level spike encoding can create excessive event volume.

Proposed solution

Hierarchical text encoding:

```
tokens
  ↓
semantic units
  ↓
events
  ↓
spike packets
  ↓
hippocampal memory
```

The hippocampal layer therefore operates on meaningful events rather than every token.

49. Problem 4 — DG collision

Problem

Similar memories may produce overlapping DG representations.

Proposed solution

$$L = L_{reconstruction} + \lambda L_{sparsity} + \mu L_{orthogonality}$$

plus:

- competitive inhibition,
- adaptive sparsity,
- learned dictionary,
- homeostasis.

50. Problem 5 — Teacher forcing

Problem

How does the network know which CA3 neurons should represent a new episode?

Proposed solution

Use novelty-driven allocation.

If:

$$\delta > \theta$$

then:

ACh-like signal

↓

lower threshold

↓

increase plasticity

↓

competitive CA3 allocation

Strong feedforward excitation plus lateral inhibition can create self-organized winner selection, while intrinsic plasticity can balance neuron usage.

Pasted markdown(20260807-144408)

51. Problem 6 — Single synapses cannot reliably retrieve memories

Problem

A single strongly weighted synapse does not necessarily provide enough current to cross the postsynaptic threshold.

Proposed solution

Use:

population convergence + recurrent amplification

rather than single-synapse retrieval.

This was directly observed in the early simulation work.

Pasted markdown(20260807-144408)

52. Problem 7 — CA3 runaway excitation

Problem

Strong recurrent excitation can cause:

network avalanche

that activates unrelated memories.

Proposed solution

Introduce:

- inhibitory interneurons,
- sparse recurrence,
- synaptic normalization,
- homeostatic thresholds,
- attractor-specific inhibition,
- neuromodulatory retrieval gating.

53. Problem 8 — Catastrophic forgetting

Problem

Continuous STDP can overwrite older memories.

Proposed solution

Dual memory:

$$Fast\ CA3 \rightarrow Replay \rightarrow Slow\ cortical\ memory$$

plus:

- replay,
- synaptic consolidation,
- importance-weighted plasticity,
- homeostasis.

54. Problem 9 — Synaptic saturation

Repeated presentation can drive:

$$w \rightarrow w_{max}$$

The specification estimates saturation under repeated STDP updates.

SPIKING_MEMORY_SPEC

Proposed solution

Use:

```
weight normalization
soft bounds
homeostatic plasticity
metaplasticity
```

55. Problem 10 — False attractors

Two memories may accidentally share enough recurrent structure to create an unintended attractor.

Proposed solution

Measure:

$$P(\text{false retrieval})$$

and reduce it through:

- DG separation,
- sparse CA3 assemblies,
- stronger inhibition,
- orthogonalization,
- synaptic normalization.

56. Problem 11 — Temporal jitter

Real enterprise streams are not perfectly synchronized.

Suppose:

```
SQL arrives at 0 ms
Graph arrives at 4.7 ms
Text arrives at 5.2 ms
```

rather than exact times.

Proposed solution

Use delay distributions rather than a single delay:

$$d \sim P(d|relation)$$

and learn optimal synaptic delays.

Learning delays alongside weights has been identified as a potential extension for temporal coding.

Pasted markdown(6)

57. Problem 12 — Missing modality

Suppose:

```
SQL = available
Graph = unavailable
Text = unavailable
```

The memory should still work.

Therefore:

$$M = \{SQL\}$$

must be sufficient to activate the appropriate CA3 attractor.

Likewise:

$$Graph \rightarrow Memory$$

and:

$$Text \rightarrow Memory$$

should work independently.

This becomes a core cross-modal retrieval benchmark.

58. Problem 13 — Contradictory modalities

Suppose:

SQL:

Alice $\rightarrow$ Google

Text:

"Alice moved to Microsoft."

The system must not silently merge them.

Instead:

conflict detection

↓

separate episodic states

↓

temporal/contextual resolution

If:

$$t_1 < t_2$$

then both may be valid:

2025:

Alice → Google

2026:

Alice → Microsoft

59. Problem 14 — Novel event versus noisy event

A high prediction error does not always mean a new memory.

It may mean:

sensor corruption
database corruption
malformed text
outlier

Therefore novelty should eventually become:

$$Novelty = f(prediction\ error, confidence, context, data\ quality)$$

rather than:

$$Novelty = prediction\ error$$

alone.

60. Problem 15 — Enterprise scale

Dense recurrent CA3 connectivity is not scalable.

Proposed solution

Use:

sparse recurrent connectivity
+
modular CA3
+
sharding
+

event-driven simulation
+
GPU/neuromorphic execution

The original engineering specification proposes modular sharding as a scaling mechanism.

SPIKING_MEMORY_SPEC

61. Problem 16 — Lack of advantage over vector databases

This is the most important non-biological problem.

A research system must demonstrate an actual computational advantage.

Potential target advantages are:

Property	Conventional vector memory	Proposed spiking memory
One-shot binding	Limited	Local plasticity
Partial cue completion	Similarity-dependent	Attractor dynamics
Temporal coding	Indirect	Native
Continual learning	Difficult	Local plasticity + replay
Cross-modal association	Shared embedding	Spike convergence
Neuromorphic execution	Limited	Native target
Exact enterprise query	Strong	Not intended

The system should therefore **not claim universal superiority**.

Its strongest proposed niche is associative, temporal, continual, multimodal memory rather than exact database retrieval. Earlier analysis reached the same conclusion.

Pasted markdown (2)(2)

62. Problem 17 — Biological plausibility

The architecture is inspired by hippocampal organization but is not a literal biological simulation.

The mapping is computational:

Biological component	Computational abstraction
EC	multimodal gateway / novelty
DG	sparse separation
CA3	autoassociative memory
CA1	readout
ACh	plasticity modulation
Engram	distributed memory assembly
Replay	consolidation
Synaptic delay	temporal code

The model should therefore be described as **biologically inspired**, not as a complete reproduction of hippocampal physiology.

63. Problem 18 — Current prototype retrieval shortcut

The current implementation still contains explicit familiarity/engram bookkeeping and signature mappings. `spiking_multimodal_memorynew +1`

Proposed solution

V2 must eliminate these from the retrieval path.

The benchmark should compare:

`symbolic-assisted retrieval`

against:

pure CA3 attractor retrieval

and the latter should become the principal scientific result.

64. Problem 19 — Current text encoder is insufficient

The prototype text mechanism is a spiking tokenizer rather than a complete semantic language representation.

Proposed solution

Use a hierarchical semantic encoder followed by a spike transformation.

Research question:

Does semantic similarity $\rightarrow$ neural trajectory similarity?

This remains an open research question.

65. Problem 20 — Statistical validity

A test such as:

10/10 retrieval

is insufficient to establish general capability.

The final system requires:

$$N \gg 10$$

episodes and multiple random seeds.

Report:

$$mean \pm std$$

confidence intervals and statistical significance.

66. Required Experimental Program

A research-grade evaluation should contain at least six benchmark families.

Experiment A — Encoding fidelity

Measure:

$$Error_{SQL}$$

$$Error_{Graph}$$

$$Error_{Text}$$

Experiment B — Pattern separation

Measure:

$$J_{input}$$

versus:

$$J_{DG}$$

Expected:

$$J_{DG} < J_{input}$$

Experiment C — Pattern completion

Cue fractions:

- 10%
- 20%
- 40%
- 60%
- 80%

Measure:

$$Overlap = \frac{|Retrieved \cap Target|}{|Retrieved \cup Target|}$$

Experiment D — Multimodal retrieval

Evaluate:

SQL → Graph
 SQL → Text
 Graph → SQL
 Graph → Text
 Text → SQL
 Text → Graph

Experiment E — Continual learning

Measure:

$$Retention(N)$$

as new memories arrive.

Experiment F — Ablation

Remove one mechanism at a time:

- DG
- STDP
- recurrent CA3
- delays
- ACh
- homeostasis
- replay

Measure the change in:

Accuracy, Capacity, Energy, Latency

67. Core Research Questions

The complete program can be reduced to the following questions.

RQ1

Can heterogeneous modalities be bound into one episodic memory using spike timing rather than a shared embedding?

RQ2

Does DG sparsification improve multimodal memory separation?

RQ3

Can CA3 recurrent dynamics perform genuine pattern completion without symbolic retrieval?

RQ4

Does polychronous temporal coding increase associative capacity?

RQ5

Can local STDP provide stable one-shot multimodal memory formation?

RQ6

≡ **ChatGPT** ✓ Prevent catastrophic forgetting under continuous multimodal input   ...

RQ7

Does the architecture provide measurable advantages over vector/graph memory under temporal and continual-learning workloads?

These questions are consistent with the previously defined research program around minimal spiking topology, capacity/interference, spike-based encoding, replay consolidation, and neuromorphic scaling. Pasted markdown(5)

68. Future Work

68.1 Learned Synaptic Delays

Instead of:

$$d_{ij} = \text{constant}$$

learn:

$$d_{ij} \leftarrow d_{ij} + \Delta d_{ij}$$

so that the network can discover useful temporal relationships.

68.2 Adaptive Polychronous Groups

The current theory assumes a fixed temporal resolution.

Future work should determine:

$$C = f(N, k, D, \delta t)$$

empirically.

68.3 Multi-timescale Memory

Introduce:

milliseconds:
synaptic events

seconds:
working memory

minutes/hours:
episodic state

days:
consolidated memory

68.4 Cortical Predictive Memory

A neocortical module can learn:

$$P(E_{t+1}|E_t)$$

while CA3 stores:

$$E_t$$

This would allow the system to move from memory retrieval toward prediction.

68.5 Sequence Attractors

Instead of:

$$M_1$$

use:

$$M_1 \rightarrow M_2 \rightarrow M_3$$

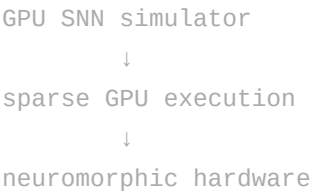
as a stable temporal trajectory.

This could support event forecasting and sequential reasoning.

68.6 Neuromorphic Implementation

The ultimate computational target is event-driven hardware.

Potential implementations include:



The original research program explicitly identifies neuromorphic hardware and efficient GPU simulation as scaling directions.

Pasted markdown(5)

69. Final Proposed System

The complete system can therefore be summarized as:

$$\begin{aligned} D_{SQL} &\rightarrow E_{SQL} \\ D_{Graph} &\rightarrow E_{Graph} \\ D_{Text} &\rightarrow E_{Text} \end{aligned}$$

followed by:

$$E_{SQL}, E_{Graph}, E_{Text} \rightarrow EC \rightarrow DG \rightarrow CA3 \rightarrow CA1$$

with:

EC → novelty

DG → pattern separation

CA3 → pattern completion

STDP → association

delays → temporal binding

polychronization → multimodal coincidence

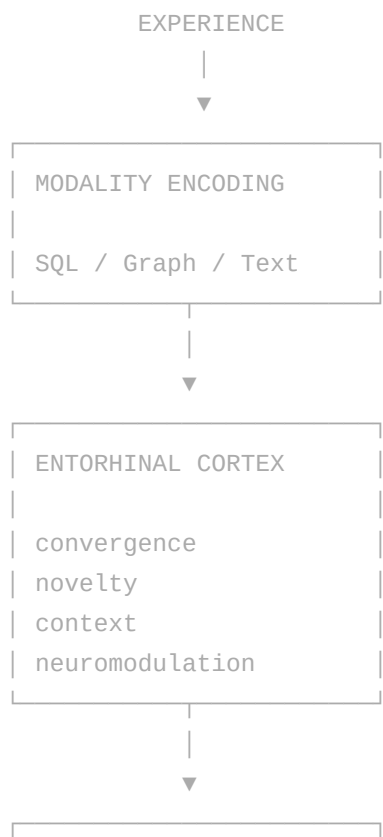
engram → distributed episodic representation

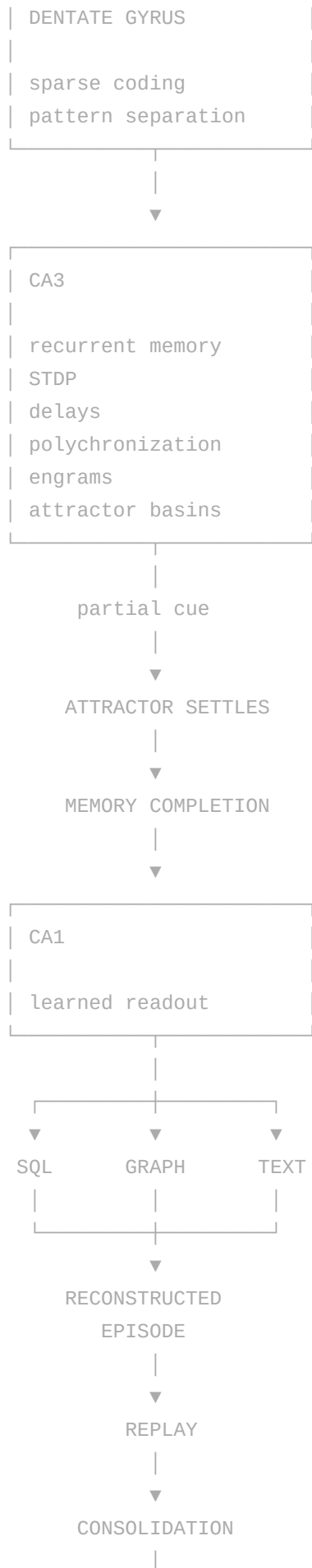
CA1 → multimodal reconstruction

and:

replay + slow learning → long – termconsolidation

The resulting computational cycle is:





70. Scientific Position of the System

The strongest formulation of the proposed contribution is therefore **not**:

“A spiking replacement for SQL, vector databases, or LLMs.”

It is:

A polychronous, multimodal, hippocampal-inspired associative memory substrate in which heterogeneous structured, relational, and sequential information is transformed into sparse spatiotemporal spike patterns and bound through local plasticity and recurrent attractor dynamics.

The central scientific hypothesis is:

spatial sparsity + temporal precision + recurrent association + local plasticity

The critical experimental transition from the current prototype to the research-grade system is the removal of symbolic retrieval assistance and demonstration that **partial SQL, graph, or text cues independently drive CA3 toward the same multimodal attractor through neural dynamics alone**. The current implementation already establishes the basic EC→DG→CA3→CA1 pathway and several component-level behaviors, but its reported 10/10 completion and 0.91 engram stability should be treated as prototype evidence rather than final evidence for the full hypothesis.

spiking_multimodal_memorynew

The resulting research program is consequently centered on a single question:

Can a sparse, delay-heterogeneous, locally plastic spiking network become a v

That question, rather than enterprise retrieval performance alone, defines the core-AI and computational-neuroscience contribution of the proposed architecture.

Pasted markdown(5)

Sources



ChatGPT can make mistakes. Check important info.

+ Ask anything

